

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING COURSE
CODE: CSA4305

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs

such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I Deepak.V-192411021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in dark ink. The signature is written in a cursive style and appears to read "Deepak".

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application	20	Correctly	Applies	Limited or	No or

of Knowledge		applies concepts to solve the question/problem	concepts with minor mistakes	partially correct application	incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response

1) Responsive Web Design:

Name: Deepak V
Reg No: 192411021
Course Code: CBA4305
Course Name: Internet
Programming CO2-AT3

Responsive Web Design (RWD) is a web development approach that enables a website to automatically adjust its layout, images, and content according to different screen sizes. It ensures user-friendly browsing experience regardless of the device they use.

Importance:

- It provides a consistent user experience across all devices.
- Improves website accessibility and readability.
- Enhances Search Engine Optimization (SEO).
- Reduces website maintenance cost.
- Increases user engagement and reduces bounce rates.

CSS Techniques:

1) Flexible Layouts:

Use percentage-based widths instead of fixed pixel values.

2) Media Queries:

```
@media (max-width: 768px) {
```

```
  .container {
```

```
    flex-direction: column;
```

```
  }
```

3) Flexible Images:

```
img {
```

```
  max-width
```

```
  height: auto;
```

```
}
```

Improvements:

- Adopt a mobile-first design approach
- Optimize images for faster loading
- Use responsive navigation menus.
- Test the website on multiple browsers.

2) Client-Side Validation Using Javascript:

Client-side validation is the process of checking user input in the web browser before the data is submitted to the server. It ensures that users enter valid and correctly formatted information, reducing invalid submissions and improving overall user experience.

Purpose:

- Prevents submission of invalid data.
- Reduce server requests
- Improves user experience
- Ensures proper data format.
- Reduces the chances of user input errors.

Javascript Event Handling:

- Onsubmit → Validates entire form
- oninput → Validates data of user
- onchange → Checks input after value changes
- onblur → Validates when input field loses focus.

Role of Regular expression:

```
< form onsubmit = "return validate ()">  
< input type = "email" id = "email" placeholder  
< input type = "tel" id = "phone" Placeholder = "Phone" >  
< button > Submit < / button >  
< / form >  
  
< script >  
function validate ()  
return / ^ [ ^ \s @ . ] + @ [ ^ \s @ ] + \. [ ^ \s @ + $ % /  
    . test email . value ) & # / { 0 - 9 } { 10 } $ /  
    test ( phone . value );  
}  
< / script >
```

Improvement:

- Displaying real-time error messages below input fields.
- Using HTML5 validation attributes such as required and pattern.

3) CSS Box Model and Responsive Layout:

The CSS Box Model is a fundamental concept in web development that defines how elements are displayed and spaced on a webpage. Every HTML element is considered a rectangular box consisting of 4 components: ~~content~~ content, padding, border and margin.

Components of CSS Box Model:

- Content: the actual text, image, or other information displayed inside the element.
- Padding: The space between the content and the border, providing internal spacing.
- Border: A line that surrounds the content and padding.
- A Margin: The space outside the border that separates the element from other elements.

CSS Positioning Techniques:

- Static: Default positioning where elements appear in the normal document flow.
- Relative: Position an element relative to its normal position while maintaining its original space.
- Absolute: Positions an element relative to the ~~browser~~ nearest positioned ancestor.
- Fixed: Positions an element relative to browser.
- Flexbox: Creates flexible one-dimensional layouts.
- Grid: Creates two-dimensional layouts.

Causes of Layout Breaking:

- > Using fixed-width elements instead of flexible layouts.
- > Excessive use of absolute positioning.

4) Document Object Model (DOM) and Dynamic Content Updates:

The Document Object Model (DOM) is a programming interface for HTML and XML documents. It represents a webpage as a tree-like structure, where each HTML element is treated as an object or node.

Role of DOM:

- Accessing and modifying HTML elements
- Updating web page content
- Handling user events
- Changing CSS styles dynamically
- Interactive web applications
- Improving Responsiveness

Javascript DOM Manipulation:

- getElementById() - Selects element using ID
- getElementByClassName() - Selects using class
- querySelector() - Selects first matching CSS selector.
- querySelectorAll() - Selects all matching CSS selector.
- InnerHTML - Changes HTML content of element
- textContent - updates only the text inside an element
- Style - Dynamically changes CSS properties.

Improvements:

- > Minimize ^{unnecessary} DOM Manipulation
- > Update only required elements.
- > use efficient DOM selection methods.
- > load scripts asynchronously.